Consequa-X: An Evidence and Intervention Protocol for Consequential Agent Actions

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***Abstract*—An agent's selected operation, tool response, reported assessment and external effect can disagree. A refund may commit before an acknowledgement is lost, and another actor may satisfy the requested state. Consequa-X records these distinctions in portable evidence traces, verifies temporal contracts from independent readback and executes isolated intervention replays with explicit unknown, invalid and budget-exhausted outcomes. The contribution is a software and evidence protocol; its contrastive and permutation Shapley estimators adapt existing methods. Audited synthetic studies retain 1,280 original calibration trials, 384 fresh shared-draw comparisons and 2,048 repeated physical sampling trials. Several original nominal 95% intervals undercovered specified effects. Conservative alternatives retain uncertainty at homogeneous samples but have substantial precision and detection costs. Physical refund enumeration shows different initial-action and consequence interactions under an explicit on-reach positional game. A separate 24-episode local-model validation found six contradicted status claims among ten assessable claims for one prompt, while another abstained in every structured status but retained affirmative language in some rationales. The reusable core supports explicit application verifiers and separates successive effect witnesses in its evidence graph. These results establish auditable implementation behavior and measured limitations in the declared settings; they do not establish a new estimator or broad attribution superiority.**

***Index Terms*—agent systems, outcome verification, evidence provenance, counterfactual replay, software artifacts, uncertainty**

I. Introduction

A consequential tool operation has distinct stages: selection, dispatch, execution, response, reported assessment and external observation. A service can commit an effect before its acknowledgement is lost. Retrying can create another effect unless the logical operation's identity and idempotency semantics are retained. Conversely, a successful response can coexist with an incomplete or incorrect effect. Explaining such runs requires more than reading the last tool response or the agent's prose.

State satisfaction and effect origin also differ. An external refund may satisfy the requested balance or count while remaining unlinked to this agent. A source-supported origin link is a provenance claim; it does not establish counterfactual necessity. A replay effect is a separate claim about a declared intervention, continuation policy and outcome distribution.

Consequa-X makes these distinctions inspectable. Its core contains typed records, schema validation, application verification and evidence-only explanation. Its research adapters provide isolated refund worlds, public-only policy inputs, a local Ollama adapter and bounded replay experiments. Reported assessments are captured outputs, not measurements of hidden reasoning.

This paper contributes (1) a reproducible protocol linking operation identity, attempts, responses, assessments, independent readback, conclusions and origin; (2) replay infrastructure with declared games, actual-call accounting and complete refusal records; and (3) audited evidence exposing uncertainty and reporting limits in specified synthetic and exploratory local-model settings. It presents a protocol/software artifact rather than a superiority comparison.

II. Related Work

Causal Agent Replay (CAR) performs same-policy interventions on recorded decisions, evaluates user-specified outcomes and uses contrastive suffix and permutation Shapley attribution [1]. Consequa-X implements an independent binary adaptation with its own typed-world, evidence and refusal boundaries. Applying a verified outcome to replay is therefore not an estimator novelty claim. Policy-Conditioned Counterfactual Credit studies changes in final verified success under specified interventions and a continuation policy [2]; its learning objective differs from post-hoc external-effect diagnosis, but verified targets and valid interventions are shared concerns.

CausalLoss-Fin jointly intervenes on agent decisions and repairable infrastructure messages [3]. Its deterministic programmatic evaluation limits transfer to stochastic language policies, but it already occupies the broad combination of financial effects and joint decision/fault attribution. A future comparison must expose applicable variables to every method rather than charge omitted infrastructure effects to decisions.

Verified Tool Calls distinguishes acknowledgements from independently checked effects and uses unknown postcondition conclusions [4]. Callability Is Not Operability studies controlled interface ambiguity and terminal agent claims [5]. Three-valued verification and response/effect separation are foundations and comparison requirements here. These targeted primary-source checks do not constitute a systematic review or an independent reproduction of those papers' empirical results.

III. Evidence Protocol and Core Interfaces

A. Objects, contracts and trust

Let H be visible history, A a logical action, E an execution attempt, R its response, B a timestamped reported assessment and O an independent readback observation. A contract C declares payload, count, deadline and stability horizon. The verifier conclusion $V(C,O)$ is satisfied, violated or unknown. Origin is separately confirmed, other actor or

unknown. Evaluator-only simulator truth never becomes a policy or explainer input.

The built-in refund predicate uses integer currency minor units. An eligible effect matches payment, amount and currency, commits in the declared time window and persists through the horizon. Exact-count contracts also constrain the count during the stability interval. A stale watermark or inadequate scope and completeness yields unknown rather than a confident absence claim. Source authority, scope, completeness and watermark declarations remain trusted adapter inputs. Schemas and hashes validate structure and changed bytes, not source authenticity or real transactional guarantees.

B. Reusable application boundary

Applications can supply an explicit read-only verifier through `verify(action, observation, verifier=...)`. The wrapper checks receipt type, contract identity, observation identity, source, watermark and timing. Domain predicate semantics, authentication and read-only adapter behavior are the application's responsibility. There is no evaluation of arbitrary predicate strings. An offline artifact-delivery example constructs captured records and checks independently readback content digests without importing the refund simulator or executing writes.

`explain(trace)` reads a captured verification and performs no replay. Refund target matching is built in; other predicates receive a neutral explanation unless a trusted domain matcher explicitly qualifies attributable effects against payload and temporal rules. Matchers can return only unique IDs within the receipt's attributable IDs. Unknown receipts are not interpreted as target matches. A qualified provenance link still is not a causal effect.

C. Successive witnesses and graph meanings

Repeated readbacks can disagree about the same effect identifier. The graph therefore retains observation-specific witnesses and an identity-only shared effect node. Dispatch-to-effect edges attach to the witness supplying origin metadata; verification edges use witnesses in that receipt's own observations. A later conflicting actor, corrected payload or reversal cannot overwrite an earlier witness and inherit its unqualified origin link. Shared identifiers do not authenticate or reconcile the records. Estimated-influence edges require captured event identifiers and an explicitly supplied intervention estimate.

IV. Replay Targets, Games and Uncertainty

A. Targets and conditional suffix effects

The refund adapter exposes three targets: initial action selection, global state-contract satisfaction and action-linked consequence. A known global violation is false for the linked target. Satisfied state with exclusively external qualifying effects is false for that target; insufficient provenance can instead yield unknown. The linked target combines global satisfaction and qualifying operation-linked witnesses rather than reconstructing an agent-only world. Unknown observations do not acquire numeric labels.

For known binary target Z, set bad outcome $X = 1 - Z$. Prefix replay holds factual decisions before position k and resamples the suffix. With one observed factual bad label x_f, the conditional estimand is

$$\gamma_k = \mathbb{E}[X \mid \text{prefix held, suffix resampled at } k] - x_f. \quad (1)$$

The factual term is one observed constant, not a population rate estimate. Negative values indicate rescue relative to that observation. An earlier irrelevant position can have a nonzero suffix effect through a later stochastic choice. A latest nominal rescue is selected only when every planned position is estimable and its interval supports a negative effect. This selection has no simultaneous coverage guarantee and does not establish a unique cause.

B. Held-decision coalition games

Let N be reached factual positions. Coalition S requests holding captured decisions at positions in S; other decisions are sampled. Reached held decisions must match the typed public stage. In the *strict* game, an earlier termination leaving a requested held position unreachable invalidates the draw. Finite samples avoiding invalidity do not define the population game.

The separate *on-reach* game treats requested positions beyond a normal termination as inactive and uses the actual completed episode. Reached incompatible operations remain invalid; provider failures and exhausted budgets are not normal inactivity. This is an explicit alternative game, not a repair of retained strict results.

For a defined game $v(S) = \mathbb{E}[X \mid S]$, signed Shapley values allocate the held-decision effect [6]:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \quad (2)$$

Fresh forward/reverse permutation walks form one paired contribution vector. Independent empty/full draws diagnose the endpoint scale. Values are signed allocations under the declared game, not normalized responsibility shares. Positive values mean holding the factual decision increases bad outcomes; negative values mean it is protective.

C. Intervals and refusals

Contrastive defaults use exact binomial Clopper–Pearson bounds translated by $-x_f$ [7]. Signed pair observations lie in $[-1, 1]$; the default Hoeffding half-width for P independent pairs and failure allowance α is [8]

$$h = \sqrt{\frac{2\log(2/\alpha)}{P}}. \quad (3)$$

This is the half-width before clipping the interval to $[-1, 1]$. Endpoint differences use two exact binomial marginals with a split failure allowance. Each marginal has confidence $1 - \alpha/2$; the difference interval is $[L_f - U_e, U_f - L_e]$. The union bound requires valid marginal bounds, not independence between the two arms. Bootstrap and normal intervals remain explicit legacy comparators. These are standard conditional methods with sampling assumptions and precision costs, not

distribution-free guarantees for dependent model continuations.

Each replay starts a fresh declared world. Held replies consume no fresh calls; every attempted fresh call, including a provider/protocol failure, consumes budget. Every planned branch remains recorded. Unknown, invalid, failed and unstarted branches are not dropped or redrawn; affected estimates are refused. Targets and interval variants reuse captured evidence without extra policy calls.

D. Analytic refund reference

Consider the declared on-reach refund mechanism with adequate budget, complete horizon-covering readback and, for the linked target, sufficient provenance. The planted factual decisions are selection s, retry r and final stop t. Fresh selection occurs with probability p and fresh timeout retry with probability q under the declared independent choice law. The first attempt commits despite its timeout; without idempotency, a retry commits a second refund. Exactly one effect therefore occurs with probability $p(1-q)$. For the consequence bad-outcome game, the four relevant coalition values are

$$\begin{aligned} v(\varnothing) &= 1-p(1-q), & v(\{s\}) &= q, \\ v(\{r\}) &= 1, & v(\{s,r\}) &= 1. \end{aligned} \tag{4}$$

The final stop is a dummy: adding t changes none of these values. Averaging the two orders of the active players gives

$$\begin{aligned} \phi_s &= -\tfrac{1}{2}(1-p)(1-q), \\ \phi_r &= \tfrac{1}{2}(1+p)(1-q), \qquad \phi_t = 0. \end{aligned} \tag{5}$$

Their sum is the endpoint difference $p(1-q)$; the selection/retry interaction $v(\{s,r\})-v(\{s\})-v(\{r\})+v(\varnothing)$ is $(1-p)(1-q)$. At $p=q=0.5$, these are shares $[-0.125, 0.375, 0]$, endpoint 0.25 and interaction 0.25. With idempotency, any selected refund creates one effect, so consequence shares become $[-(1-p), 0, 0]$, as for initial action selection; the interaction is zero. These are population references derived from the declared mechanism, not fitted to sampled outcomes. A call cap can refuse the finite estimate despite a defined ideal reference. Unknown evidence or strict unreachable-held decisions do not acquire numeric references from these forms.

A standalone standard-library check uses exact rational arithmetic on a five-by-five (p,q) grid, two idempotency settings and two targets: 100 control cases. It checks all coalition values and six permutation orders against these forms without importing the framework or reading sampled study data. This checks the stated mathematical law, not simulator execution or sampling validity.

V. Audited Evaluation

A. Original calibration and finite analysis

The original development study retained 1,280 trials across ten specified mechanisms, with 128 repetitions. Of 223,232 planned branches, 198,690 completed, 4,062 were invalid, 256 failed at the deliberate cap and 20,224 were unstarted.

TABLE I
Original nominal 95% coverage at the protective stop.

Retry q	Contrastive covered	Shapley covered
0.1	100/128	107/128
0.5	122/128	121/128
0.9	103/128	103/128

TABLE II
Fresh shared-draw coverage: 32 trials per row.

q	Bootstrap	Exact binomial	Normal	Hoeffding
0.1	26/32	31/32	23/32	32/32
0.5	31/32	32/32	32/32	32/32
0.9	31/32	32/32	29/32	32/32

Standalone audit scripts checked raw evidence, seeded outcomes, statistics and accounting. Table I reports a preselected protective-stop consequence position with 128 available, reference-eligible estimates in each row.

Separate finite-distribution calculations yielded protective-stop normal pair coverage of 83.7775% at $q = 0.1, 0.9$ and 92.6377% at $q = 0.5$ for 16 pairs. For interior Bernoulli q, homogeneous draws force an empirical percentile bootstrap interval to a point, giving coverage ceiling $1-q^K-(1-q)^K$. At $K = 16$ and $q = 0.1, 0.9$, this is about 81.47%. These findings concern the specified adaptation; they do not establish coverage of upstream CAR implementations.

B. Shared-draw conservative comparison

A separately frozen follow-up retained 384 trials across twelve cases, 66,048 planned branches and 87,054 actual scripted calls. Variants reused the same points, draws, seeds and refusals. Exact probability weights over sample counts 8,16,32,64 and eleven probabilities yielded minimum grid coverage 7.72% for the limiting empirical percentile bootstrap, 85.15% for Wilson and 96.68% for Clopper–Pearson. The bootstrap grid is its infinite-resample limit, distinct from the replay's finite seeded bootstrap.

Table II is the protective-stop timeout position, not every case. The full summary includes exact-binomial coverage 30/32 for a retry-pivot case. Finite observed coverage can fall below nominal despite a conservative population method. At $q = 0.1$, mean normal versus Hoeffding width was about 0.180 versus 1.358; at $q = 0.5$, 0.344 versus 1.184. Hoeffding detected the negative reference sign in 0/32 trials at both weaker settings, and 32/32 at $q = 0.9$. Wider intervals therefore impose a substantial detection cost.

The hardened independent audit passed 3,283,523 assertions at its recorded derived-artifact inventory; twenty targeted corruptions were rejected. A separate SciPy numerical audit covered 1,290 binomial cases and 1,290 endpoint differences, with maximum binomial endpoint disagreement approximately 2.66×10^{-14}. SciPy was an audit dependency, not a framework runtime dependency. Finite grids and synthetic replays do not establish live-model sampling independence or simultaneous attribution coverage.

C. Physical interactions and variable length

An independent physical enumeration covered four deterministic fresh-choice components, eight held masks, two games and eight cases, retaining 512 planned branches and 718 actual scripted calls. Outcomes came from ordinary payment readback rather than injected payoff values. With selection and timeout retry probabilities $p = q = 0.5$, unsafe retry and exactly-one verification, on-reach consequence shares were $[-0.125, 0.375, 0]$, with selection/retry interaction 0.25. Action-selection shares were $[-0.5, 0, 0]$. Idempotency changed consequence shares to $[-0.5, 0, 0]$ and removed the interaction. This is a declared programmatic mixture, not observed model interaction recovery.

The strict population game remained undefined at interior probabilities; missing provenance, lag and small budgets retained their distinct limits. The independent audit passed 27,277 checks; twenty-nine semantic corruptions were rejected. The original export manifest and data remain preserved alongside the derived canonical inventory. Two sampler integration controls exercised 32 pairs and 32 endpoint draws, with eighteen coefficients within a broad 0.3 regression tolerance. That tolerance alone is not a calibration result.

D. Repeated physical sampling

A separately frozen protocol exercised the actual permutation sampler across all eight declared physical mechanisms, both games and 128 fresh repetitions per mechanism/game: 2,048 trials. Each trial planted the same three factual decisions and used 32 forward/reverse permutation pairs, one draw per coalition evaluation and 32 independent draws per endpoint, yielding 320 planned branches. The planted factual path was specified rather than selected by seed search. The cap was 1,000 actual calls including three factual calls, except for the four-call budget control. Collection used normal intervals; Hoeffding and exact-binomial endpoint differences were rescored from identical draws without additional calls, altered points or redrawn refusals. The installed checkpoint, precollection protocol, all compressed trial/branch rows and replicate-zero full evidence with separately stored evaluator truth are retained.

The run retained all 655,360 branches: 536,150 completed, 37,423 invalid, 256 failed and 81,531 unstarted. It consumed 628,535 scripted calls, including 6,144 factual calls, and zero language-model calls. On-reach execution recorded 37,392 branches with inactive requested holds; these are normal-termination records, distinct from strict invalidity and budget exhaustion.

A standalone auditor importing no framework implementation passed 44,698,925 assertions over the retained trials, branches, 4,210 complete traces and all summary rows. Its receipt binds the frozen source revision and original 52-file manifest. Derived tables retain every row separately from that sealed collection; auditing made no generation calls.

Table III reports every declared mechanism for one specified target and position, including unavailable controls. At $(p, q) = (0.5, 0.5)$, the shared retry estimate had bias -0.00269 and RMSE 0.0611. Normal coverage 118/128 has a 95% Wilson interval $[0.862, 0.957]$; 128/128 has $[0.971, 1]$, not proof of perfect population coverage. The conservative intervals sharply reduced weak-effect sign detection. For the consequence stop dummy, normal false nonzero flags ranged from 0 to 14/128 across the six assessable mechanisms; Hoeffding flags were 0/128 in each. The deterministic zero-reference case yielded 128 normal point intervals; their correctness in that case does not validate point intervals for stochastic outcomes.

Endpoint behavior was separately assessed. For the on-reach unsafe $(0.2, 0.8)$ global-consequence endpoint, normal coverage was 92/128 (Wilson $[0.635, 0.789]$), with 35 point intervals and mean width 0.114. Exact-marginal differences covered 128/128, with mean width 0.319. These finite results demonstrate the declared precision tradeoff; they do not establish simultaneous coverage or calibration under live-model dependence.

In the on-reach game, every lagged-consequence coefficient was refused as unknown; every budget coefficient was refused after unstarted slots, despite the latter's defined ideal reference. Removing provenance preserved global-consequence availability but refused all 128 linked-consequence coefficient vectors. All nondeterministic strict coefficient vectors were unavailable and their population games undefined; the fully deterministic strict control remained defined. One finite strict interaction point at $(0.8, 0.2)$ did not make that population interaction defined. One strict linked-consequence endpoint without provenance was numeric but reference-ineligible, giving coverage 0/0 rather than an accuracy result. Pooled-coalition interaction diagnostics have no confidence interval and supply no coverage or sign-detection result. The full summary retains separate availability, reference eligibility, bias/RMSE, coverage uncertainty, width, flags, endpoints, interactions and refusal rows for every target, position, game and interval variant. Reused targets and variants are not independent replications.

E. Exploratory local-model validation

A frozen installed Qwen3 4B checkpoint completed six validation cases, two repetitions and two unchanged prompt profiles, using 59 actual calls within a 72-call cap. No episode failed or remained unstarted. Model identity, prompt/payload hashes, public traces and the exact source checkpoint are retained. The original and revised profiles induced different action sequences; this is not a pure reporting experiment or an attribution-method comparison.

The revised profile's conditional contradiction rate is undefined, not zero error. In the other-actor case, revised-profile readback satisfied state while origin was external and action linkage false. In the expired-request case, the original profile correctly described ineligibility but issued a refund; its failure assessment contradicted achieved effect. Effect satisfaction and eligibility-policy correctness are distinct targets.

A separate unblinded annotation pass within the AI-assisted development workflow marked affirmative processing clauses in 10/12 original and 7/12 revised final rationales. Five of those seven revised rationales also hedged independent verification. A statement that a second attempt committed can be literally

TABLE III

ALL EIGHT DECLARED CASES: ON-REACH RETRY COEFFICIENT FOR GLOBAL CONSEQUENCE, 128 PLANNED REPETITIONS PER ROW. N/H DENOTES NORMAL/HOEFFDING. AVAILABILITY A AND POPULATION-REFERENCE ELIGIBILITY R EACH HAVE DENOMINATOR 128. FLAGS DENOTE CORRECT DIRECTIONAL DETECTION FOR NONZERO REFERENCES AND FALSE NONZERO FLAGS FOR ZERO REFERENCES. DASHES DENOTE UNDEFINED QUANTITIES; 0/0 DENOTES AN UNDEFINED COVERAGE RATE.

Case (p, q) or control	Reference	A ; R	Covered N / H	Mean width N / H	Flags N / H
Unsafe $(0.5, 0.5)$	0.375	128 ; 128	118/128 ; 128/128	0.226 / 0.960	128/128 ; 8/128
Unsafe $(0.2, 0.8)$	0.120	128 ; 128	124/128 ; 128/128	0.154 / 0.960	123/128 ; 0/128
Unsafe $(0.8, 0.2)$	0.720	128 ; 128	123/128 ; 128/128	0.218 / 0.773	128/128 ; 128/128
Deterministic $(1, 1)$	0	128 ; 128	128/128 ; 128/128	0 / 0.960	0/128 ; 0/128
Idempotent $(0.5, 0.5)$	0	128 ; 128	123/128 ; 128/128	0.244 / 0.960	5/128 ; 0/128
No provenance $(0.5, 0.5)$	0.375	128 ; 128	120/128 ; 128/128	0.228 / 0.960	128/128 ; 7/128
Lagged readback $(0.5, 0.5)$	—	0 ; 0	0/0 ; 0/0	—	—
Four-call cap $(0.5, 0.5)$	0.375	0 ; 128	0/0 ; 0/0	—	—

TABLE IV

LOCAL-MODEL VALIDATION WITH EXPLICIT CLAIM DENOMINATORS.

Quantity	Original v1	Revised v2
Completed/planned episodes	12/12	12/12
Actual calls/profile cap	36/36	23/36
Definite final status claims	12/12	0/12
Contradictions/assessable claims	6/10	0/0
State satisfied/violated/unknown	6/4/2	7/3/2
Linked satisfied/false/unknown	6/4/2	5/5/2

true while two effects violate an exactly-one task. These annotations are descriptive; no independent human verification or inter-rater reliability has been established. Unknown structured fields do not ensure prose abstention. The saved auditor passed 2,227 assertions and rejected seven corrupted smoke copies without additional generation. Fixed profile order, tiny sample size, shared fault conditions and an exposed validation family limit interpretation. The held-out reserve was not generated.

VI. DISCUSSION AND LIMITATIONS

The evidence protocol supports different, inspectable claims about responses, state, origin and intervention effects. Its refusal rules prevent unsupported numerical outputs but can sharply reduce availability. Availability can depend on interventions and must remain visible beside numerical results. Conservative intervals do not make undefined games numeric, resolve missing evidence, or remove the need to assess the actual sampling law.

The environment is synthetic and narrow. The artifact-delivery example validates an integration interface, not a second production-domain experiment. Declared source authority and watermarks are not authenticated transactional guarantees. The model experiments use one local model and exposed development/validation families; they do not estimate broad model performance or held-out transfer. State and linked targets sharing outcomes are not independent replications. Claims about hidden reasoning cannot be inferred from reported assessments.

Future superiority research requires faithful modular baselines with matched evidence, target definitions, candidate variables, snapshots and actual budgets. Joint context/fault attribution must expose infrastructure variables. Independent template and fault-composition families, predeclared precision, replay-fidelity checks and cluster-aware uncertainty are needed for generalization claims. Those comparisons are future work, not completed evidence of this paper.

VII. CONCLUSION AND ARTIFACT AVAILABILITY

Consequa-X provides a reusable evidence core and auditable replay adapters for separating reported assessments, external outcomes, origin and intervention effects. The retained studies expose uncertainty failure, precision costs, variable-length game restrictions and prose/structured-status disagreement. The useful result is a reproducible protocol/software artifact with measured limits, without claiming a new estimator or general method superiority.

Stable version 1.0.0 is publicly available under MIT on PyPI [9]. Its wheel and source distribution were independently downloaded and hash-checked, and an isolated registry installation was verified on October 15, 2026. The wheel contains the runtime, schemas and typing marker; the source distribution also includes documentation, offline examples, tests and audit scripts. Private research outputs and model weights are excluded. The maintainer-controlled repository at https://github.com/Devrajsinh-Jhala/consequa-x is private. Git history, retained protocols, raw evidence and evaluator-only truth are retained there for authorized collaborators; this paper does not assert unrestricted access to that research archive. Each historical study retains its own source revision and manifest; later engineering fixes do not silently replace earlier data. A public immutable research archive and review-access arrangements have not been established. Framework imports and offline examples require no model credentials or required runtime dependencies. All reported payments are simulated; model weights are identified rather than redistributed.

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